Perfect! Let's carefully identify **all the suitable stored procedures** that would make this database fully functional and support all 5 modules you mentioned. I'll structure them according to the entities and modules, keeping in mind operations like insert, update, delete, retrieval, and calculations.

1️⃣ Categories

- `sp_AddCategory(category_brand, model_name, product_type, screen_size, color, supplier_id)`
- `sp_UpdateCategory(category_id, category_brand, model_name, product_type, screen_size, color, supplier_id)`
- `sp_DeleteCategory(category_id)`
- `sp_GetCategoryDetails(category_id)`
- `sp_ListAllCategories()`

2️⃣ Suppliers

- `sp_AddSupplier(supplier_name, contact_no, email, warehouse_address, bank_ac_no)`
- `sp_UpdateSupplier(supplier_id, supplier_name, contact_no, email, warehouse_address, bank_ac_no)`
- `sp_DeleteSupplier(supplier_id)`
- `sp_GetSupplierDetails(supplier_id)`
- `sp_ListAllSuppliers()`

3️⃣ Products

- `sp_AddProduct(product_name, category_id, supplier_id, screen_size, color, processor, ram, primary_storage, secondary_storage, gpu, cost_price, wholesale_price, sale_price, quantity)`
- `sp_UpdateProduct(product_id, ...)` (update any product attribute and auto-log changes in `products_history`)
- `sp_DeleteProduct(product_id)`
- `sp_GetProductDetails(product_id)`
- `sp_ListProductsByCategory(category_id)`
- `sp_ListAllProducts()`

4️⃣ Staff

- `sp_AddStaff(staff_name, role_id, location_id, staff_contact, staff_email, staff_cnic, staff_bank_acno, staff_salary, staff_hiring_date)`
- `sp_UpdateStaff(staff_id, ...)`
- `sp_DeleteStaff(staff_id)`

- `sp_GetStaffDetails(staff_id)`
- `sp_ListStaffByRole(role_id)`
- `sp_ListStaffByLocation(location_id)`

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5 Orders & Order Items

- `sp_CreateOrder(customer_id, order_type, location_id, payment_method, delivery_method, shipping_fee, discount_amount, items_json)`
  (items_json = list of product_id, qty, unit_price)
- `sp_UpdateOrder(order_id, status, payment_method, shipping_fee, delivery_method)`
- `sp_DeleteOrder(order_id)`
- `sp_GetOrderDetails(order_id)`
- `sp_ListOrdersByCustomer(customer_id)`
- `sp_ListOrdersByDate(start_date, end_date)`
- `sp_AddOrderItem(order_id, product_id, qty, unit_price)`
- `sp_UpdateOrderItem(item_id, qty, unit_price)`
- `sp_DeleteOrderItem(item_id)`

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6 Returns & Return Items

- `sp_CreateReturn(order_id, customer_id, location_id, items_json, return_reason, notes)`
- `sp_UpdateReturn(return_id, return_reason, notes)`
- `sp_DeleteReturn(return_id)`
- `sp_GetReturnDetails(return_id)`
- `sp_AddReturnItem(return_id, item_id, qty_returned, notes)`
- `sp_UpdateReturnItem(ri_id, qty_returned, notes)`
- `sp_DeleteReturnItem(ri_id)`

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7 Locations

- `sp_AddLocation(location_name, location_type, address, contact_no, managed_by, staff_capacity)`
- `sp_UpdateLocation(location_id, ...)`
- `sp_DeleteLocation(location_id)`
- `sp_GetLocationDetails(location_id)`
- `sp_ListLocationsByType(location_type)`

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8 Stocks & Inventory

- `sp_AddStock(product_id, location_id, qty_stocked, unit_cost, manager_id)`
- `sp_UpdateStock(stock_id, qty_stocked, unit_cost)`

- `sp_DeleteStock(stock_id)`
- `sp_GetStockDetails(stock_id)`
- `sp_ListStocksByLocation(location_id)`
- `sp_ListStocksByProduct(product_id)`
- `sp_UpdateInventory(product_id)` *(calculate total_qty from stocks, update inventory_amount, auto-log in `inventory_history`)*
- `sp_GetInventory(product_id)`
- `sp_ListAllInventory()`

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9 Purchases

- `sp_AddPurchase(supplier_id, product_id, quantity, unit_cost, received_by, location_id, remarks)`
- `sp_UpdatePurchase(purchase_id, quantity, unit_cost, remarks)`
- `sp_DeletePurchase(purchase_id)`
- `sp_GetPurchaseDetails(purchase_id)`
- `sp_ListPurchasesBySupplier(supplier_id)`
- `sp_ListPurchasesByDate(start_date, end_date)`

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10 Stock Movement

- `sp_MoveStock(product_id, from_location_id, to_location_id, quantity_moved, unit_cost, approved_by)`
- `sp_UpdateStockMovement(smr_id, quantity_moved, unit_cost)`
- `sp_DeleteStockMovement(smr_id)`
- `sp_GetStockMovement(smr_id)`
- `sp_ListStockMovementsByProduct(product_id)`
- `sp_ListStockMovementsByLocation(location_id)`

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1 1 Invoices & Invoice Items

- `sp_CreateInvoice(order_id, customer_id, customer_name, customer_contact, customer_email, payment_method, delivery_method, discount_amount, location_id, items_json)`
- `sp_UpdateInvoice(invoice_id, payment_method, delivery_method, discount_amount, notes)`
- `sp_DeleteInvoice(invoice_id)`
- `sp_GetInvoiceDetails(invoice_id)`
- `sp_ListInvoicesByCustomer(customer_id)`
- `sp_ListInvoicesByDate(start_date, end_date)`

1️⃣2️⃣ Sales & Sales History

- `sp AddSale(order id, order item id, return id, return item id, product id, quantity, cost_price, wholesale_price, sales_price, amount, location_id, created_by)`
- `sp_UpdateSale(sales_id, quantity, amount)` *(auto-log in `sales_history`)*
- `sp_DeleteSale(sales_id)`
- `sp_GetSaleDetails(sales_id)`
- `sp_ListSalesByDate(start_date, end_date)`
- `sp_ListSalesByLocation(location_id)`
- `sp_ListSalesByProduct(product_id)`

1️⃣3️⃣ Roles

- `sp_AddRole(role_name, description)`
- `sp_UpdateRole(role_id, role_name, description)`
- `sp_DeleteRole(role_id)`
- `sp_ListRoles()`

✅ Notes:

- Many procedures are **auto-logging changes** into their respective history tables (`products_history`, `inventory_history`, `stocks_history`, `sales_history`, etc.).
- For POS, `sp_CreateOrder` **and** `sp_CreateInvoice` can be combined into a workflow stored procedure that updates `orders`, `order_items`, `invoices`, `stocks`, and `inventory` simultaneously in a **transaction** to maintain consistency.
- Audit and report queries can rely on these procedures for summarization (`total sales`, `profit/loss`, `inventory valuation`, etc.).